Melis Can

AI/ML Engineer | Software Engineer

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AI/ML engineer with hands-on experience building end-to-end machine learning systems and multi-agent LLM applications, from data pipeline design to model deployment. Strong foundation in explainable AI, with a track record of turning research ideas into working software. Interested in roles that combine technical depth with real-world product impact.

WORK EXPERIENCE

AI Engineer Intern, *Bluesense AI* — 06/2025 – 08/2025, Istanbul, Türkiye

- Built and trained a **MobileNetV2-based CNN** model for facial skin analysis, achieving **95% macro F1-score** across 10 classes to drive high-accuracy product features.
- Applied data preprocessing, feature extraction, and augmentation techniques to **improve model robustness and generalization** for the SmartBeauty mobile application, ensuring a reliable user experience.

IT Intern, *ADM Electricity Distribution* — 07/2024 – 08/2024, Denizli, Türkiye

- Designed and executed **SQL** queries for data extraction and transformation to build reporting workflows that support data-driven **project management** decisions.
- Analyzed operational datasets to identify trends and inefficiencies, contributing to **system optimization** and better **strategic planning**.

EDUCATION

B.Sc. in Software Engineering, *Istanbul Atlas University (Full Scholarship)* — 10/2022 – 06/2026, Istanbul, Türkiye

- **HUAWEI Student Developers (HSD),** *Core Team Member of Technical Development and Training Committee:* Collaborating with a multidisciplinary team to organize technical workshops.

PROJECTS

FinAgent: Multi-Agent AI Investment & Portfolio Platform — 04/2026 – Present

- Developed a comprehensive investment analysis platform for Turkish retail investors; designed a proprietary Inflation Protection Score (IPS, 0-100) as a core UX metric and integrated real-time market data (USD/TRY, Gold, BIST 100) via CBRT EVDS API and Yahoo Finance.
- Architected a robust FastAPI backend featuring an 8-agent orchestrator with LLM support for Turkish natural language QA, alongside a Monte Carlo simulation engine executing 1,000 GBM scenarios.
- Built a dynamic portfolio simulator utilizing Next.js 14 that automatically recommends asset allocations based on user risk profiles and delivers interactive data visualizations.
- Engineered and optimized the analytical data infrastructure to execute nominal/real return comparisons, backtesting, and inflation/policy rate correlation analyses using 6 years of historical data (2020-2026).

Technologies Used: Next.js 14, FastAPI, LLM Orchestration, Python, Monte Carlo (GBM), EVDS API, Yahoo Finance API, Tailwind CSS, Interactive Data Visualization

Multi-Floor Indoor Navigation with Semantic Search and LLM Integration, *Graduation Project* — 01/2026 – 05/2026

- Developed a fully implemented multi-floor indoor navigation system tested on a real six-floor building; designed a modular pipeline resolving free-form Turkish/English queries into optimal routes via a web application and CLI.
- Implemented 7 pathfinding algorithms (A*, Weighted A*, Theta*, UCS, BFS, IDA*, Bidirectional A*) on a 65k-cell ASCII grid with a time-calibrated cost model using asymmetric stair/elevator costs derived from real walking speeds and MET values.
- Built a three-tier query resolver against 168 named POIs across 31 space types; integrated LLaMA-3.3-70B via Groq API for NL query parsing and route narration, with deterministic fallback ensuring full navigation availability when the API is unavailable.
- Delivered a Flask + vanilla JavaScript SPA with Navigation, Explore, and Floor Plans modes; implemented server-side route rendering via Pillow producing annotated per-floor PNG overlays on scaled floor plan images.

Tech Stack: Python, Flask, Vanilla JS, LLaMA-3.3-70B, Pillow/ImageDraw, REST API, JSON

Explainable AI-Driven Clinical Decision Support for Genetic Variant Classification 02/2026 – Present

- Developed an end-to-end clinical decision support system utilizing XGBoost and LightGBM to assist healthcare professionals in genomic variant classification, effectively bridging the gap between complex AI predictions and clinical actions.
- Performed advanced feature engineering on genomic tabular datasets and integrated SHAP-based explainability analysis to provide transparent, clinically verifiable model insights.
- Evaluated model performance across key metrics (Accuracy, F1-Score, ROC-AUC), strictly optimizing hyper-parameters to minimize false negatives to ensure patient safety and clinical reliability.

Tech Stack: Python, XGBoost, LightGBM, SHAP, Scikit-learn, Tabular Data Engineering

AI-Powered Real-Time Industrial Voice Assistant 11/2025 – 01/2026

- Designed and implemented an advanced speech-to-text and RAG pipeline integrating Whisper (ASR) and Llama-3 to provide low-latency, context-aware troubleshooting support for industrial environments.
- Developed a robust intent detection system combining semantic retrieval (SentenceTransformers, FAISS) with pattern matching to accurately interpret user commands and operational workflows.
- Built an interactive Streamlit interface featuring Text-to-Speech (TTS) integration, achieving an 80% Exact Code Match Accuracy while diagnosing acoustic background noise factors to optimize hands-free usability.

Tech Stack: Whisper ASR, Llama-3, SentenceTransformers, FAISS, Streamlit, Python, NLP

TECHNICAL SKILLS

AI & LLM Systems: LLM Orchestration (LangChain, LangGraph), Multi-Agent Systems, Agentic Function Calling & Tool Use, RAG, Whisper ASR, SentenceTransformers, FAISS

Languages & Backend: Python, Java, SQL, FastAPI, WebSocket

ML / DL Frameworks: PyTorch, TensorFlow/Keras, Scikit-learn, Classical ML Models (Tree-based, Linear, Ensemble Methods), Dimensionality Reduction, XGBoost, LightGBM, SHAP, Pandas, NumPy, Model Evaluation (F1-score, ROC-AUC)

Frontend & Web Tools: Next.js 14, React, Tailwind CSS, Streamlit, Leaflet.js

Developer Tools & DevOps: Git, Docker, Vercel, Cursor, Claude Code

TRAININGS & PROGRAMS

AI Scholar, *Artificial Intelligence & Technology Academy* 12/2025 – Present

- Enrolled in a Google Türkiye-supported AI program focused on project-based learning and hackathon-based evaluation.

Future Talent Program, *Citi Foundation, YGA & UP School* 02/2026 – 04/2026

- AI and Big Data acceleration program, focusing on sustainability analytics and automation.

Data Science Trainee, *EPAM Systems* 11/2025 – 02/2026

- Developed end-to-end **ML pipelines** including comprehensive EDA, supervised/unsupervised learning, **hyperparameter optimization**, and **MLOps workflows** on complex tabular datasets.

LANGUAGES & CERTIFICATIONS

Turkish: Native

English: Professional Working Proficiency

Convolutional Neural Networks (DeepLearning. AI)

Developing Generative AI Applications using Python (IBM)

Foundations of Project Management (Google)

Forward Program (McKinsey.org)